3.1.3 Current Electricity

- **Charge, current and potential difference**

Electric current as the rate of flow of charge; potential difference as work done per unit charge.

$$I = \frac{\Delta Q}{\Delta t} \text{ and } V = \frac{W}{Q}$$

Resistance is defined by $R = \dfrac{V}{I}$

- **Current/voltage characteristics**

For an ohmic conductor, a semiconductor diode and a filament lamp; candidates should have experience of the use of a current sensor and a voltage sensor with a data logger to capture data from which to determine $I - V$ curves.

Ohm's law as a special case where $I \propto V$

- ## Resistivity

$$\rho = \frac{RA}{L}$$

Description of the qualitative effect of temperature on the resistance of metal conductors and thermistors. Applications (e.g. temperature sensors).

Superconductivity as a property of certain materials which have zero resistivity at and below a critical temperature which depends on the material. Applications (e.g. very strong electromagnets, power cables).

- ## Circuits

Resistors in series; $R = R_1 + R_2 + R_3 + \ldots$

Resistors in parallel;

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots$$

energy $E = I\,V\,t$, $P = IV$, $P = I^2R$; application, e.g. Understanding of high current requirement for a starter motor in a motor car.

Conservation of charge and energy in simple dc circuits.

The relationships between currents, voltages and resistances in series and parallel circuits, including cells in series and identical cells in parallel.

Questions will not be set which require the use of simultaneous equations to calculate currents or potential differences.

- ## Potential divider

The potential divider used to supply variable pd e.g. application as an audio 'volume' control.

Examples should include the use of variable resistors, thermistors and L.D.R.'s. The use of the potentiometer as a measuring instrument is not required.

- ## Electromotive force and internal resistance

$$\varepsilon = \frac{E}{Q} \qquad \varepsilon = I\,(R + r)$$

Applications; e.g. low internal resistance for a car battery.

- ## Alternating currents

Sinusoidal voltages and currents only; root mean square, peak and peak-to-peak values for sinusoidal waveforms only.

$$I_{\mathrm{rms}} = \frac{I_0}{\sqrt{2}} \qquad V_{\mathrm{rms}} = \frac{V_0}{\sqrt{2}}$$

Application to calculation of mains electricity peak and peak-to-peak voltage values.

- ## Oscilloscope

Use of an oscilloscope as a dc and ac voltmeter, to measure time intervals and frequencies and to display a.c. waveforms. No details of the structure of the instrument is required but familiarity with the operation of the controls is expected.